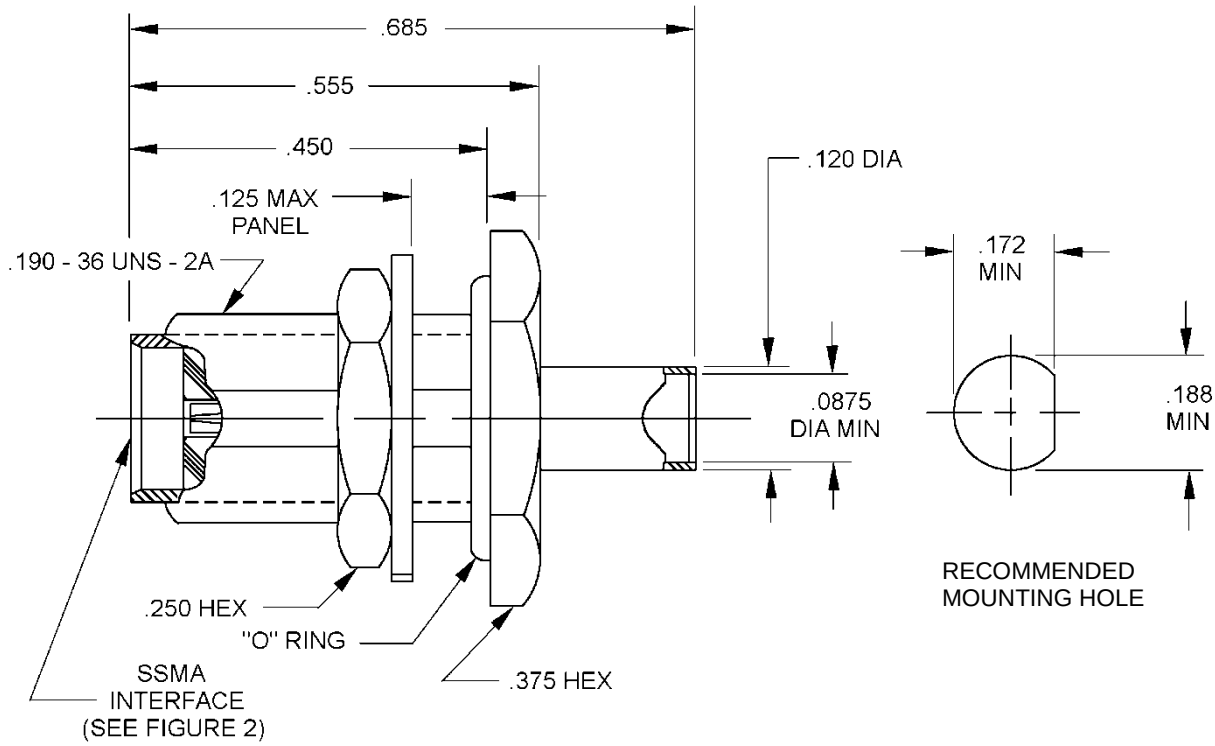
INCH-POUND
MIL-PRF-39012/138A
28 January 2013
SUPERSEDING
MIL-PRF-39012/138
27 June 1989

PERFORMANCE SPECIFICATION SHEET

CONNECTOR, RECEPTACLE, ELECTRICAL, SERIES SSMA, SOCKET CONTACT, FOR SEMIRIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Mating dimensions in accordance with MIL-STD-348.
4. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28.
5. All undimensioned pictorial representations are for reference only.
6. Dimension .685 (17.40 mm) defines the maximum overall length of the connector when assembled to the cable.
7. Method of the assembly to the cable shall be solder.
8. Unless otherwise specified, tolerances are ±.005(0.13 mm).

Inches	mm
.087	2.22
.120	3.04
.125	3.17
.172	4.36
.188	4.77
.190	4.82
.259	6.35
.375	9.52
.450	11.43
.555	14.09
.685	17.40

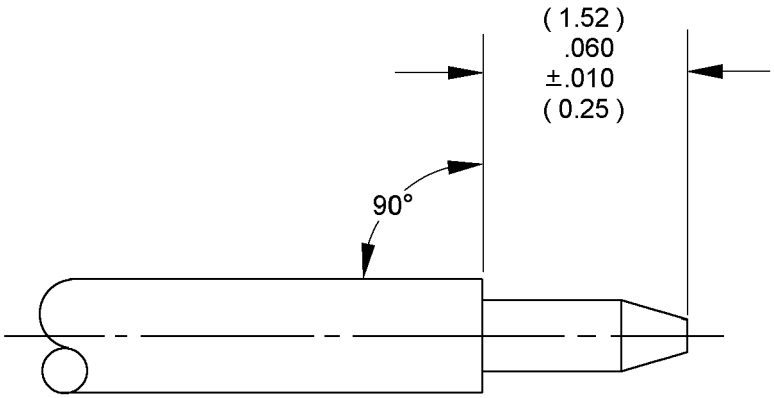
FIGURE 1. General configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:21Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number and applicable cable.

Dash no. 1/	Applicable cable 2/
Category A Field serviceable (no special tools required)	
3001	M17/133 RG405*
4001	M17/133-00001 through -00011
Category E Field serviceable (standard assembly tool kit) 3/ (see figure 2)	
3002	M17/133 RG405*
4002	M17/133-00001 through -00011

* Cable to be used when performing tests requiring cable.
1/ These connectors have captivated center contacts.
2/ MIL-DTL-17 cables are specified by the basic number. The latest version of each cable shall be applicable.
3/ All corrosion resistant steel bodied connectors shall be gold plated in accordance with ASTM B488, type 3, grade C, class 1.27,at least in the area of solder attachment.



NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Cable stripping dimensions for category E connectors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:21Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 35 GHz.

Voltage rating:
250 V rms at sea level
60 V rms at 70,000 feet

Operating temperature: -65 C to 165 C.

REQUIREMENTS:

Dimensions and configurations: see figure 1.

Force to engage and disengage;
Longitudinal force: Not applicable
Torque: 2.0 inch-pounds maximum

Mating characteristics:

Center contact (socket):
Oversized test pin: .0213 + .0001 -.0000 inch.
Test pin finish: 16 microinches
Insertion depth: .045.
Number of insertions: 1.

Insertion force test:
Steel test pin diameter: .0208 +.0001, -.0000 inch.
Insertion depth: .050 to .065.
Test pin finish: 16 microinches
Insertion force: 3 pounds maximum.

Withdrawal force test:
Steel test pin diameter: .0195 +.0000, -.0001 inches.
Insertion depth: .050 to .065.
Withdrawal force: 1 ounce.
Test pin finish: 16 microinches.

Coupling proof torque: Not applicable

Recommended mating torque: 2 inch-pounds

Hermetic seal: Not applicable

Leakage (pressurized connectors): Not applicable

Center contact retention: 4 pounds minimum, axial force.

Radial torque: Not applicable

Voltage standing wave ratio (Vswr): 1.07 + .010F (F in GHz), maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:21Z
Check the source to verify that this is the current version before use.

Moisture resistance: Method 106 of MIL-STD-202, no measurement at high humidity. Insulation resistance shall be at least 200 megaohms, within 5 minutes after removal from humidity.

Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact:	4.0	4.0
Outer contact:	2.0	Not Applicable
Outer conductor to body:	0.5	Not Applicable

Dielectric withstanding voltage: 750 Vrms, minimum at sea level.

Vibration, high frequency: In accordance with method 204 of MIL-STD-202.

Corona level:

Altitude. 70,000 feet, 190 Vrms minimum.

Shock (specified pulse): In accordance with method 213 of MIL-STD-202

Thermal shock: in accordance with method 107 of MIL-STD-202, except test high temperature shall be +85°C.

Barometric pressure (reduced): Not applicable.

RF high potential withstanding voltage: 500 Vrms minimum.
Frequency: 5 MHz
Leakage current: Not applicable.
Cable retention force: 30 pounds minimum.
Torque: 16 inch-ounces minimum.

Coupling mechanism retention force: Not applicable.

RF leakage: -90°F (F in Ghz) dB, minimum.

RF insertion loss: $.04\sqrt{F, GHz\ dB}$, maximum.

Part or Identifying Number (PIN): M39012/138- (dash number from table I).

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:21Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-PRF-39012, this document references the following:

FED-STD-H28
MIL-DTL-17
MIL-STD-202
MIL-STD-348
ASTM B488

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, AT, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5935-2012-249)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:21Z
Check the source to verify that this is the current version before use.